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[16MSCCH401]

FOURTH SEMESTER UNIVERSITY EXAMINATION, MAY - 2019

CHEMISTRY OF BIOMOLECULES

Time: 03 Hours **Maximum Marks:70**

Instructions: **I. Question 1 is compulsory, carries 10 marks.**
II. Questions 2 to 7 carry 15 marks each, answer any four.

1. Answer any FIVE. **5×2=10**

a. What is an active transport? Give two examples.
b. Give the structure of deoxy-myoglobin.
c. Explain briefly the structural features of cytochrome P-450 and its functions.
d. Define transferases with example.
e. Explain the process of denaturation in DNA and protein.
f. Write the biological mode of action of penicillin.

2. a. Write a note on essential and trace elements in biological system.
b. Give an account of chelation therapy with suitable examples. **(10+5)**

3. a. i. Draw the MO diagram of O_2 binding with heme unit.
ii. Discuss the magnetic properties of Fe and O_2 in oxy-haemoglobin.
b. What is ferrodoxin? Discuss its structural features and functions. **(10+5)**

4. a. Compare the structure property relationship among mononuclear Zn-containing enzymes.
b. Describe the catalytic cycle of peroxidase enzyme and its function. **(10+5)**

5. a. Define enzyme inhibitor and classify the inhibitors with suitable diagram.
b. Describe different methods of purification of enzymes. **(10+5)**

6. a. Explain Strecker synthesis of amino acids with mechanism.
b. With neat diagram, explain hydrogen bonding in DNA. **(10+5)**

7. a. What are antibiotics? Give the synthesis and mode of action of chloramphenical and norfloxacine.
b. Discuss the mode of action of streptomycin in biological system. **(10+5)**

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